

The code base is modularized into main.c, io.c, editor.c board.c and define.h.

- main.c will have the main control flow of the program.
- io.c handles loading board from file to the program as well as writing the board to an output file.
- editor.c mainly handle the Task3 and Task4, the editorial functions for the list and the board.
- board.c handle loading predefined board, displaying the board, menu printing functions as well as general helper functions.
- define.h is a header with function prototypes, as well as struct declaration

Ascii Art is printed at the start of the program as a welcome message.

board is defined as a global variable in main.c and declared as extern in define.h to make it accessible across all files.

board will take predefined board by default unless load_board_from_file() is called in which case will replace the default board.

How do you load a board?

load_board_from_file() function will handle loading the board if user wishes to replace the default board.

Filename is asked as user input string and it assumes the file will be on the same directory level as the code base.

fgets() is used to read the file line by line.

Newline character is removed at the end of each line.

Due to our input file format assumption of items having whitespaces in the front, we will trim these spaces so that our linkedlist will not have whitespaces which will help standardize the output.

the function will differentiate whether it is a list or an item by checking for the ":" colon delimiter.

Each list and item is dynamically allocated and linked into the doubly linked list structure.

How did you edit lists and items?

Option 3 from menu allow the editing of items within specific list. The user enter the list name, then is prompted to either add, rename, or delete items. Option 4 allows editing the board itself that is adding, renaming or deleting entire lists. Helper functions such as find_list and find_item helps the function locate the nodes by name. In the case that user doesn't add ":" for lists, it automatically check for it and add ":".

How do you save a board?

write_board_to_file() function ask for filename. The program then traverse through all the nodes until the last one, writing each list name followed by items using fprintf. The output formatting is the same as input format assumption.

What's your ASCII art about?

The ASCII art is "YEHOR VICK"- the name and nickname of two group members.

Git Repo:

https://csgitlab.ucd.ie/YehorShch/assignment_3_az_win_yehor